A new framework based on FOG Computing and Edge AI to increase the reliability of predictive maintenance of rotating machines

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***Abstract*—In today's industrial world, it is essential to ensure efficient and safe operation of machines to ensure productivity and reduce costs. Vibration analysis is one of the most effective tools used for this purpose. Vibration analysis is used to monitor the condition of rotating machines, detect potential problems early and avoid costly breakdowns. In this article, we will explore how the integration of vibration analysis with fog computing and edge artificial intelligence (Edge AI) can create a powerful system for real-time machine monitoring and predictive maintenance. Keywords—predictive maintenance, FOG computing, Edge AI, rotating machine**

I. INTRODUCTION

Management of resources in industrial computers demands strategic decision-making.
Resource constraints, heterogeneity, the dynamic and diversified nature of workloads, and the variability of FOG or Edge computing systems have made managing resources even more difficult to solve in the FOG landscape.

Recently, technologies based on artificial intelligence (AI) & machine learning (ML) have been implemented to tackle this issue. AI & ML systems that can make sequential judgments, such as reinforcement learning, appear to be the most promising for solving such challenges. However, these algorithms provide unique issues such as high variance, interpretability and online training. The dynamics of the ever-changing FOG/EDGE environment necessitate systems that learn online and adapt to an evolving computing environment [1].

II. MATERIALS AND METHODS

Integrating vibration analysis with fog computing and edge AI can create a powerful system for real-time machine tracking and maintenance prediction. Here's how these concepts can be integrated [2]:

A. *Data acquisition and collection [3]:*

- Sensors (e.g. those adapted to the context of vibration, gyroscopes, accelerometers, speed sensors) are attached to the machines so as to collect the maximum amount of data on vibrations for greater reliability.
- These sensors continuously measure vibrations to have a good real-time approach.

B. *Edge Processing (Edge AI – Edge Computing) [4] :*

- The raw vibration data is sent to nearby edge devices (e.g., industrial gateways or small computers) to improve communication performance.
- These edge devices perform initial processing and analysis using embedded AI algorithms.
- Edge AI can perform tasks like:
  - Filtering out noise (Kalman Filter).
  - Extracting relevant features from the vibration signals
  - Running fast Fourier transforms (FFT) to analyse frequency components in the frequential domain.

C. *Fog Layer [5] :*

- The fog computing layer sits between edge devices and the cloud. thus, reducing network traffic, and avoiding network bottlenecks between devices and the cloud.
- It consists of local servers or micro data centers that can aggregate data from multiple edge devices.
- In this layer, more complex analysis can be performed, such as:
  - Comparing vibration patterns across multiple machines
  - Running more sophisticated machine learning models
  - Detecting anomalies that may indicate potential failures

D. *Cloud Integration [6]:*

- While most processing occurs at the edge and fog layers, the cloud can be used for :

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- Long-term data storage
- Training of machine learning models using historical data
- Providing a centralized dashboard for monitoring multiple sites

E. Real-time Evaluation and Decision-Making [7][8]:

- The distributed nature of this system allows for real-time evaluation & analysis and fast decision-making.
- The implementation of Edge devices can immediately alert operators of critical issues.
- The fog layer can make decisions about scheduling maintenance or adjusting machine operations.

F. Predictive Maintenance [9][10] :

- By continuously analyzing vibration patterns, the system can predict potential failures before they occur.
- This allows for scheduling maintenance only when needed, reducing downtime and costs.

G. Adaptive Learning [11][12]:

- As the system collects more data, machine learning models at both the edge and fog layers can be updated to improve accuracy.
- This creates a self-improving system that becomes more effective over time.

The benefits sought behind this integration of various techniques are:

- Reduced latency: Processing at the edge means faster response times for critical issues.
- Bandwidth efficiency: Only relevant data and insights are sent to the cloud, reducing network load.
- Improved reliability: The system can continue functioning even with intermittent cloud connectivity.
- Scalability: Easy to add new machines or sensors to the network.

And the challenges to be met should allow the integration of industry 4.0 via the following considerations:

- Ensuring consistent AI model performance across distributed devices.
- Managing security in a distributed system.
- Balancing processing load between edge, fog and the other cloud layers.

III. BACKGROUND

The aim of our study is to create a novel automated framework capable of predicting and diagnosing incipient failures of rotating machinery using Edge AI. Our model has as inputs vibration signals from an accelerometer-gyroscope and has as outputs an estimation of the rotating machine condition as well as the degree of wear and fault diagnosis of the components of these machines.

First, the signals are gathered and analyzed to identify probable fault signatures. The generated signals are then analyzed to identify features associated to gearbox component defects at specified frequencies. Finally, these attributes are employed as inputs in pattern recognition models to assess whether a failure exists, in which component, and at what stage it happens.

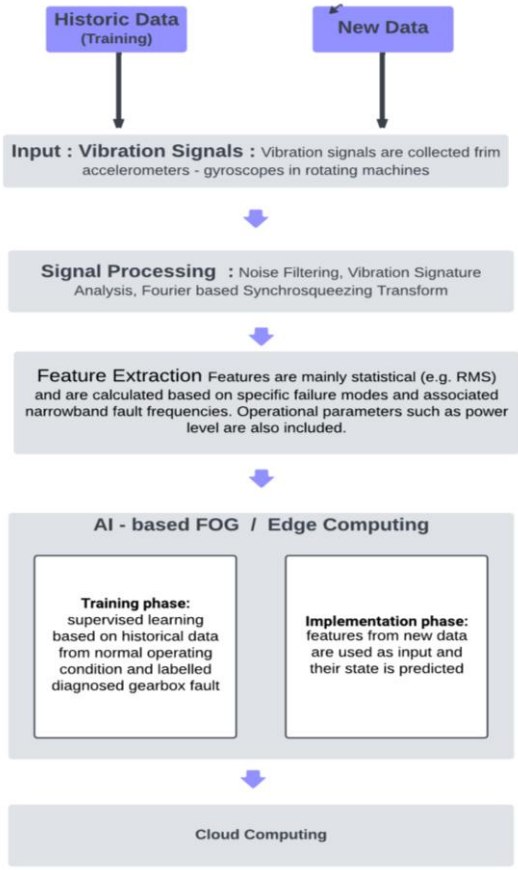


Fig 1. Methodology Flowchart

A. Noise analysis

Acquired signals often contain background noise from the environment. The most commonly used filters were Kalman filters [13], and extended Kalman filters [14] [15], but it turned out that for rotating machines TQWT techniques are more effective for denoising vibration signals from bearings and rotating machines [16]..

B. Vibration signature assessment

Induction machines in industry are defined by their signal processing method and measured using a machine vibration analysis device. ISO 2372 (108) uses the Root Mean Square (RMS) velocity of the machine housing as the criterion. According to NASA, the RMS velocity ranges from 10 to 1000 Hz. The resulting value is subject to standard limits [17].

C. The Fourier-based Synchrosqueezing Transform

The SST has two goals: to give a concentrated representation of multi-component signals in the TF plane, as well as a decomposition method for separating and demodulating the various modes. A theoretical result, initially stated in [18] in the CWT context, demonstrates its utility for well-separated low-modulated multicomponent signals. This section describes STFT-based synchrosqueezing (FSST) and concludes the approximation result of [18] to that framework. We will pay special emphasis to the differences between FSST and CWT-based synchrosqueezing (WSST) [19].

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D. AI-based fog/edge computing [1]

As IoT and machine learning usage grows, cloud and edge workloads become more diversified and dynamic. To increase human quality of life by combining fog with AI, smart resource management is necessary. Traditional cloud computing platforms rely on heuristic ways to manage resources, failing to account for diverse and dynamic workloads (20). Many of these strategies, such as the threshold-based strategy, are static heuristics set up offline for specific workload scenarios. They cannot grow applications dynamically based on workload patterns [21]. Scaling up can significantly reduce the performance of heuristic approaches. Resource contention is a significant issue in fog environments, where co-located applications fight for shared resources, leading to poor performance and SLA violations [22]. The transition from monolithic to micro-services and serverless architectures has led to greater complexity [23]. Dependency in micro-services can lead to SLO breaches due to increased communication costs and resource demands in fog computing.

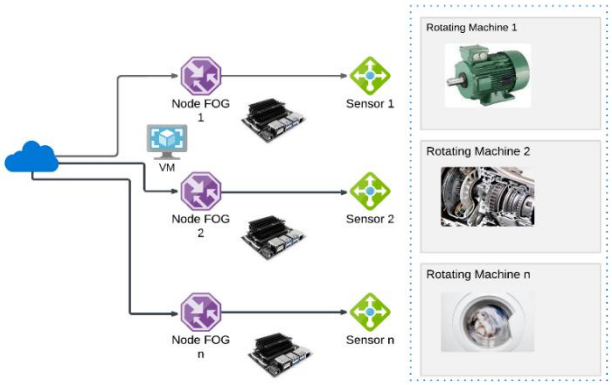


Fig. 2: Block diagram of the processing methodology

IV. The Architecture of our Framework

In this paper, the proposed framework is a new architecture, which brings together the elements of FOG Computing and also Edge Computing that are at the base of a new predictive maintenance system for rotating machines, linking it with the tasks required during its development. The complete lifecycle of the framework is illustrated in Figure 3 and consists of several steps. A data preparation process is applied on the raw time-stamped vibration data in order to generate the equivalent rotation patterns, which are then denoised in a special sub-layer of our FOG. Then, a vibration signature extraction procedure is activated, followed by an equivalent failure recognition step. The final step of the process takes place at the Cloud level, and which allows to store the obtained results, and also to feed the training step of the learning machines.

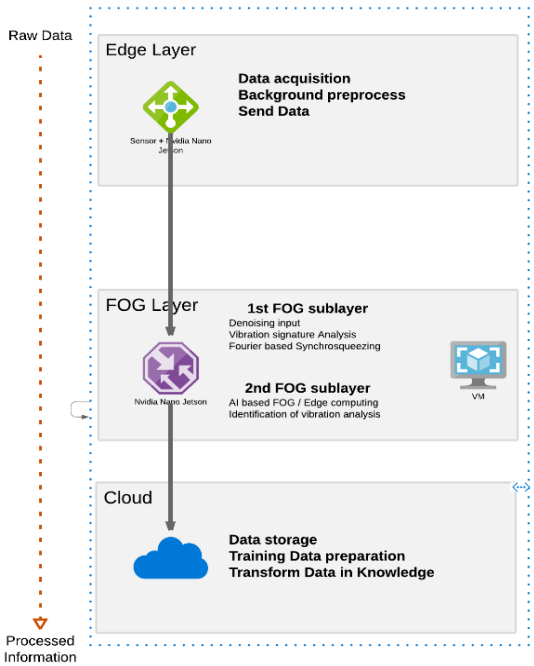


Fig. 3: Diagram of the components of our framework

V. Realisation

We will describe a new framework for predictive maintenance of rotating machinery using vibration analysis and FOG computing, based on the Jetson Nano. This framework integrates edge computing with the cloud for a complete and efficient solution.

This framework integrates edge computing with the Jetson Nano and cloud analytics to provide a powerful and cost-effective predictive maintenance solution.

This framework could significantly improve the reliability of rotating machinery while reducing maintenance costs. It would be particularly useful in manufacturing, energy or transportation industries.

The characteristics of our achievement are:

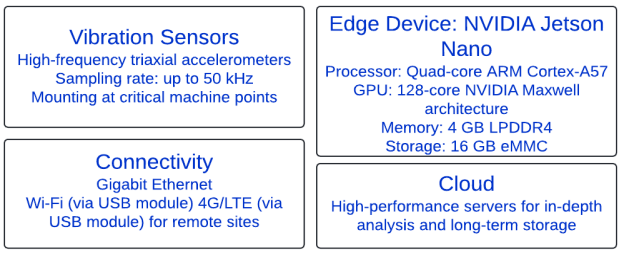


Fig. 4: Characteristics of the components of our framework

The following picture illustrates the setup of our framework for a test on a DC motor.

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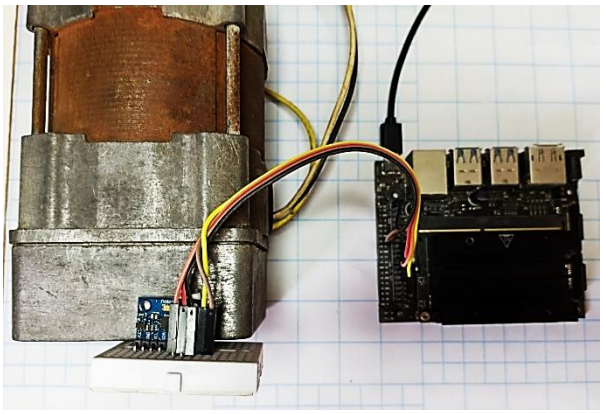


Fig. 5: Illustration of the assembly including Nvidia Nano Jetson and motor for testing framework

VI. Realisation

Here are some key points to remember:

- The Jetson Nano enables powerful local processing, reducing latency and data transmission costs.
- Real-time analytics at the edge enable rapid detection of critical anomalies.
- The cloud provides additional storage and compute capacity for deeper analytics and continuous model training.
- The architecture is adaptable to numerous types of rotating machinery.
- The system can easily scale by adding more Jetson Nanos to monitor more machines.

This framework could significantly improve the reliability of rotating machines while reducing maintenance costs. It would be particularly useful in manufacturing, energy, or transportation industries.

VII. Conclusion

The proposed framework, because of how it integrates FOG Computing and Edge AI, greatly improves the reliability of predictive maintenance for rotating machines. The methodology shows a substantial improvement to fault detection accuracy and timeliness by leveraging Edge AI's real-time processing and FOG nodes' proximity.

In particular, the distributed intelligence at the edge does reduce data latency in addition to bandwidth consumption. Consequently, this acumen allows faster anomaly detection. Edge AI models process sensor data locally, which minimizes reliance upon centralized cloud infrastructure. This increases system resilience also reduces the impact of network connectivity issues. Also, the framework is able to learn on the device continuously and this lets predictive adaptive models change with the operational traits of the machine so a stronger more accurate strategy results.

The article presents empirical results that validate the superior performance of this methodology as it can reduce costly downtime to optimize maintenance schedules regarding rotating machinery all across various industrial sectors.

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